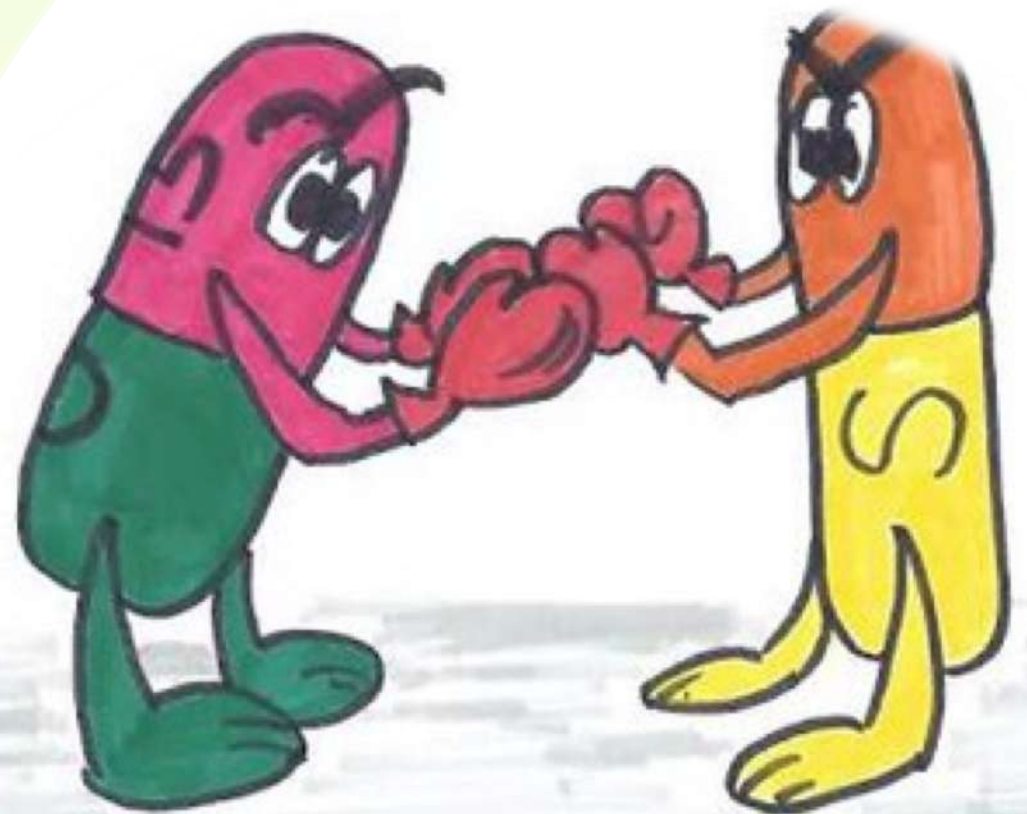


Drug-Drug interactions

Pharmacology Department



Objectives:

Predisposing factors for drug interactions

Advanced age.

Multiple medications.

Acute severe illness on top of chronic disease.

Poor renal and hepatic function.

More than one prescribing doctor.

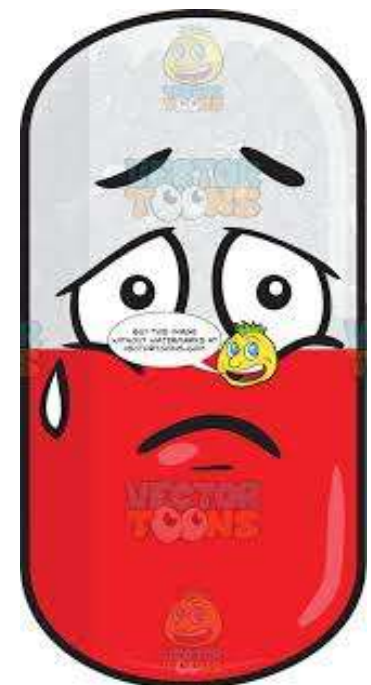


Drugs likely to be involved in interactions

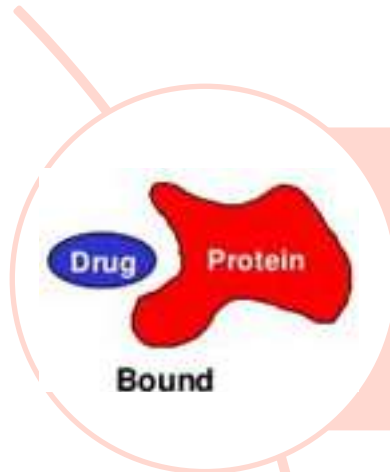
1-Drugs that are likely to **precipitate** drug interactions.



2- Drugs that are likely to be the **objects** of drug interactions



1-Drugs that are likely to precipitate drug interactions



1- Drugs that are highly protein bound, and are therefore likely to **displace object drugs from protein-binding sites**.



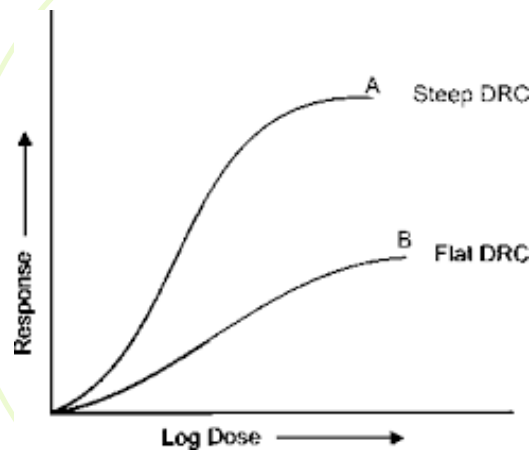
2-Drugs that alter (stimulate or inhibit) the metabolism of other drugs (**Inducers or Inhibitors**).



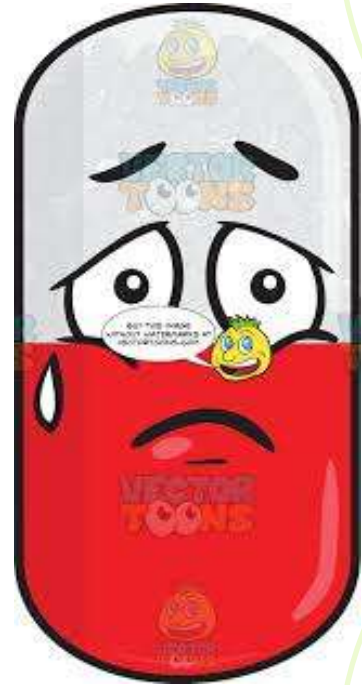
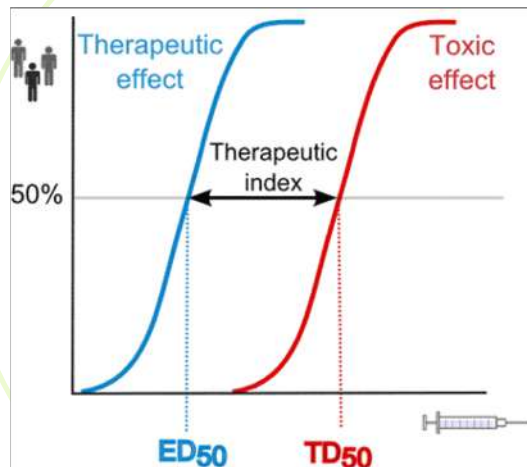
3-Drugs that affect **renal function** and alter the renal clearance of object drugs.

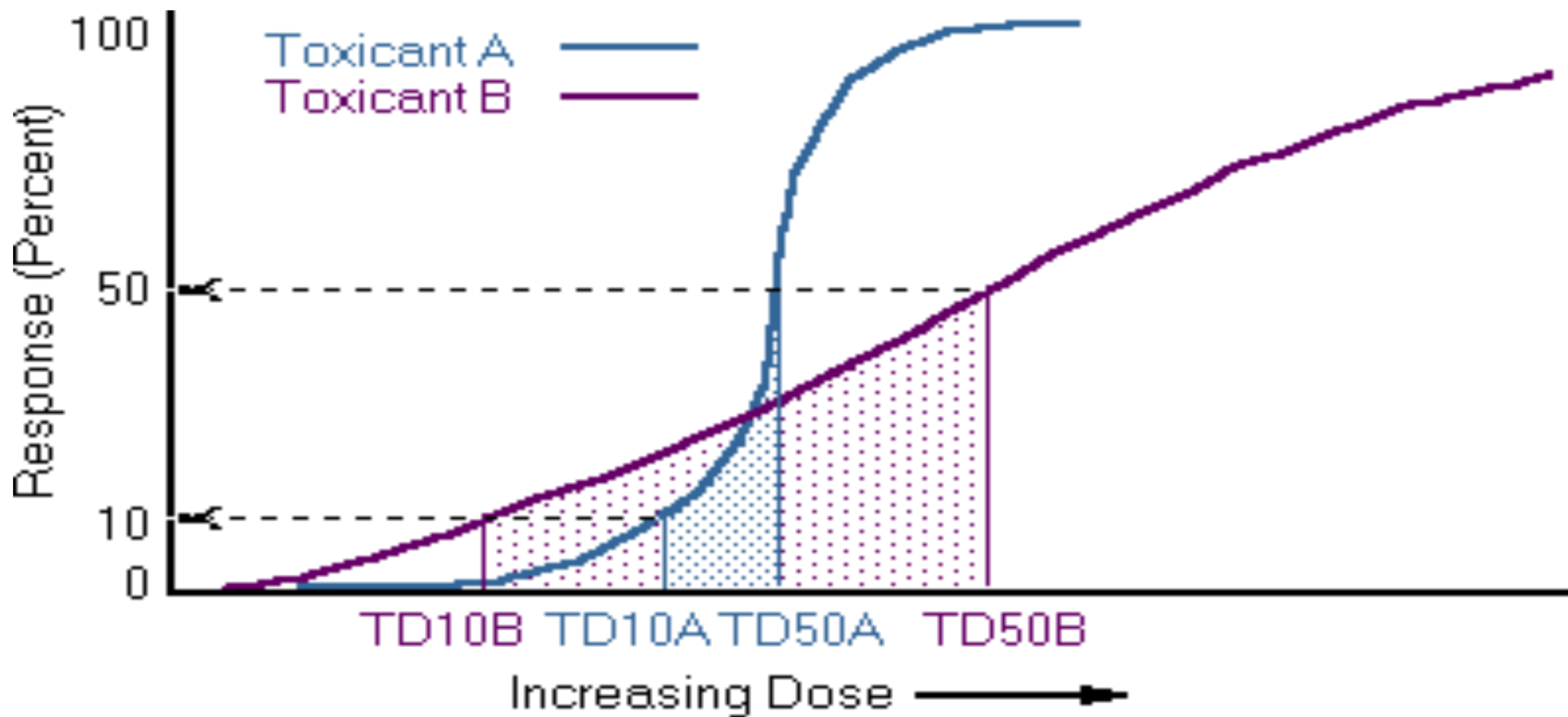
2- Drugs that are likely to be the objects of drug interactions

1-The drugs that are most likely to be the object drugs in interactions are those that have a steep dose—response curve (i.e. drugs for which a small change in dose results in a relatively large change in therapeutic effect, important in interactions causing reduced efficacy of the object drug).



2.drugs that have a low therapeutic index (i.e. drugs for which the dose at which toxic effects start to occur is little more than the therapeutic dose, important in interactions causing toxic effects of the object drug).





Knowledge of the slope is important in comparing the toxicity of various substances.

For some toxicants a small increase in dose causes a large increase in response (*toxicant A, steep slope*).

For other toxicants a much larger increase in dose is required to cause the same increase in response (*toxicant B, shallow slope*).

Identifying Potential Drug Interactions

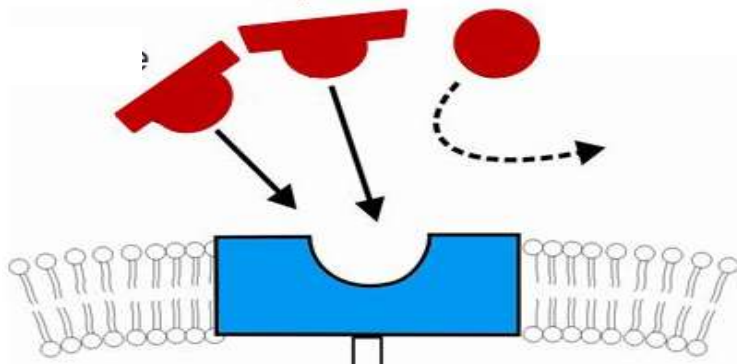
Pharmaceutical

Outside the body.
Before administration

Pharmacodynamics

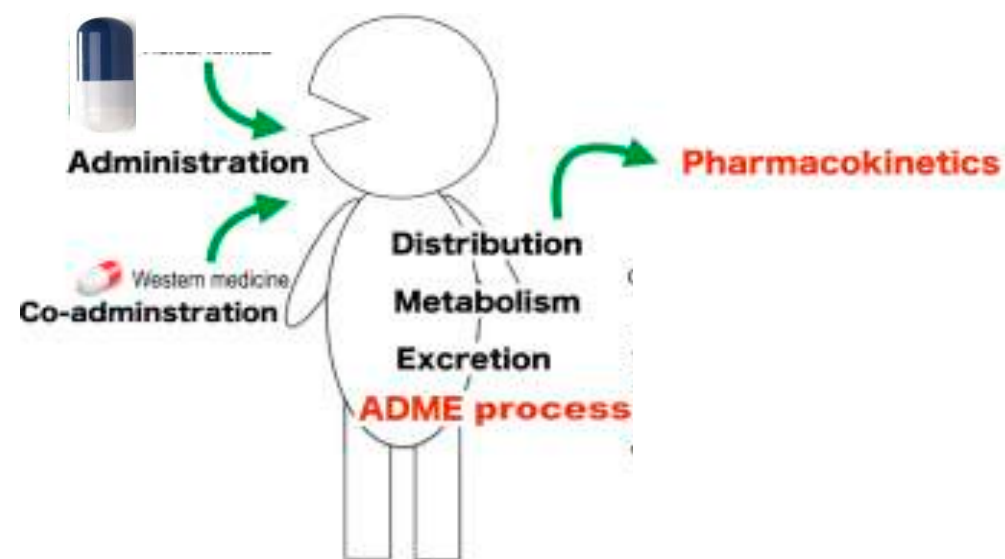
1-Interactions directly on:

- Receptors
- Body systems.



Pharmacokinetics

- 1- Interactions at the site of **absorption**
- 2- Interactions during **distribution**
 - Displacement from plasma proteins
- 3- Interactions during **metabolism**
 - Enzyme induction
 - Enzyme inhibition
- 4- interaction during **excretion**

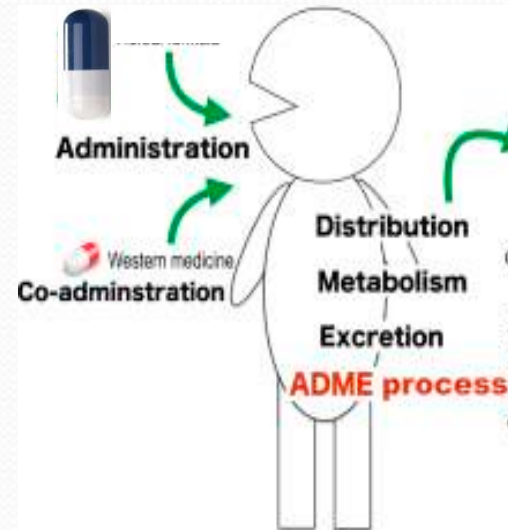


Mechanism of Drug Drug Interaction

Pharmaceutical

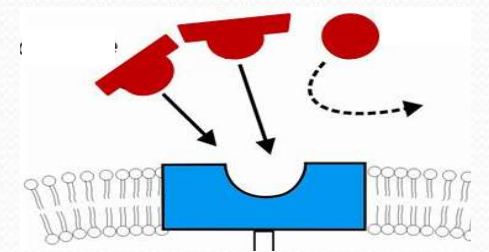
Outside the body.
Before administration

Pharmacokinetics



Pharmacodynamics

Interactions directly on:
- Receptors
- Body systems.



1- Interactions at the site of absorption

1- Gastric motility

2- GIT PH changes:

3- Binding or chelation:

4- Structure of gut wall:

2-Interactions during distribution

-Displacement from plasma proteins

3-Interactions during metabolism

- Enzyme inhibition

- Enzyme induction

Inhibition of non-microsomal enzymes

4- interaction during excretion

A- Competition for active tubular excretion.

B- Changes in urinary PH

Mechanisms of drug interactions

I- Before administration

1- Mixing drugs + IV infusion fluids:

-May change PH, solubility or activity of components.

Ampicillin, chlorpromazine and barbiturates react with dextran solution forming chemical complexes.



2- Mixing drugs prior administration:

-Protamine (Basic)+ Heparin(Acidic)---→Neutralization

-P.Z.I. + soluble insulin----→Precipitation of soluble insulin.

(reduce immediate effect of insulin)

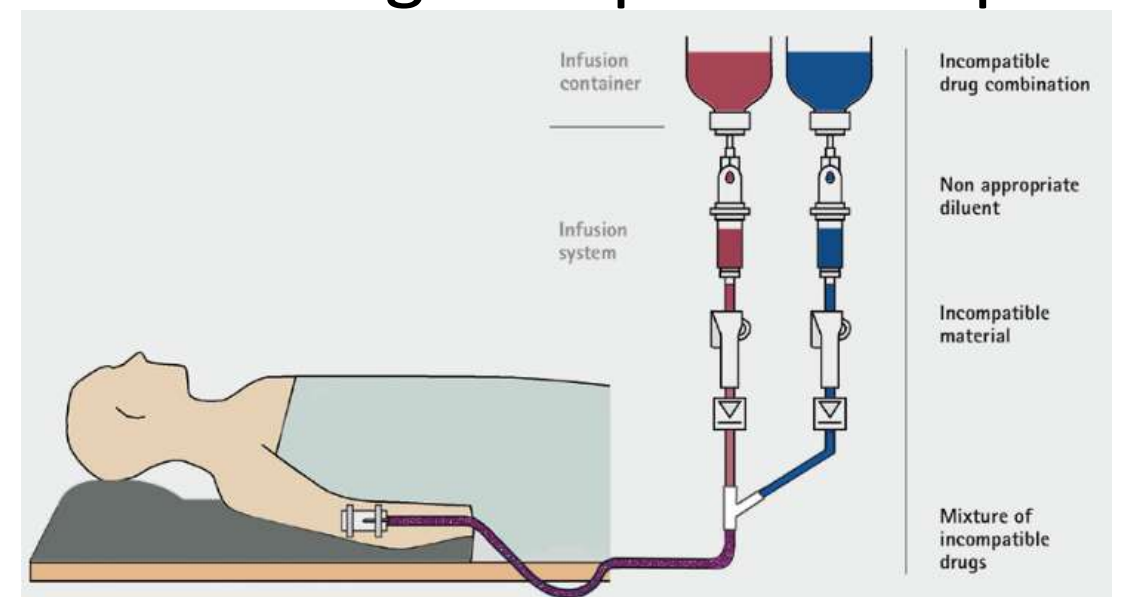
Pharmaceutical interactions

- Pharmaceutical interactions are physicochemical interactions, either:-
 - A drug with an intravenous infusion solution or
 - Two drugs in the same solution,
- resulting in the loss of activity of the drugs involved.

- **To avoid these interactions:-**

1-Give intravenous drugs by bolus injection if possible.

2-Do not add drugs to infusion solutions other than dextrose or saline.
Even in those solutions, some drugs are unstable, while others are light sensitive and must be protected from the light to prevent rapid loss of activity.



- 3- Avoid mixing drugs in the same infusion solution, unless the mixture is known to be safe (e.g. potassium chloride with insulin).
- 4- Mix the drug thoroughly in the infusion solution and check both soon after and later during the infusion for visible changes (turbidity, precipitation, or colour change).
- 5- Label all infusion bottles clearly with the name and dose of drug added and the times of starting and ending the infusion.
- 6- Use two separate infusion sites if two drugs must be infused simultaneously, unless you are sure that there is no interaction.

II- After administration

A) Pharmacokinetic interactions (ADME)

1- GIT absorption

1- Interactions
at the site of
absorption

1- Gastric
motility

2- GIT PH
changes:

3- Binding or
chelation:

4- Structure
of gut wall:

1- Gastric motility:

A- **Metoclopramide** increases gastric emptying and increases absorption of propranolol and paracetamol.

B- **Atropine** decreases gastric emptying and decreases absorption of propranolol and paracetamol.

[Absorption is mainly from the upper part of intestine]

Metoclopramide



Atropine



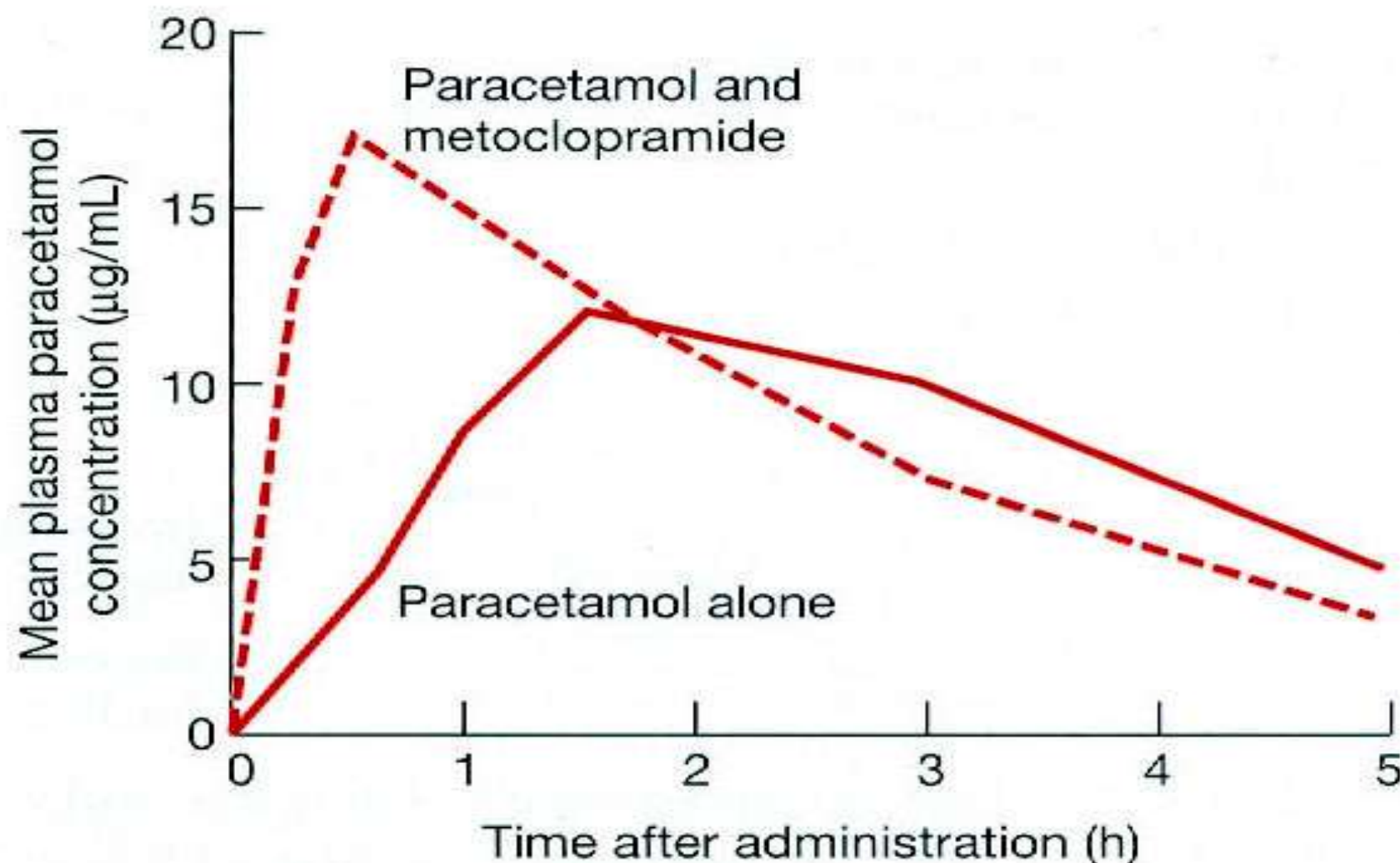


Fig. 10.1 The effect of metoclopramide on the absorption of paracetamol. When metoclopramide (10 mg i.v.) was given with paracetamol (1.5 g orally), the rate of paracetamol absorption was increased, as evidenced by a higher and earlier peak paracetamol plasma concentration. However, the extent of absorption of paracetamol was not changed, since the area under the plasma concentration versus time curve was unaffected. (Adapted from Nimmo *et al.* (1973) *Br. Med. J.* 1, 587–9, with permission.)

II- After administration

A) Pharmacokinetic interactions (ADME)

1- GIT absorption

2- GIT PH changes:

A- **Gastric acidity** increase absorption of weak acids as aspirin and phenobarbital. Ketoconazole needs acidity for absorp.

B- **Intestinal alkalinity** increases absorption of weak bases as atropine

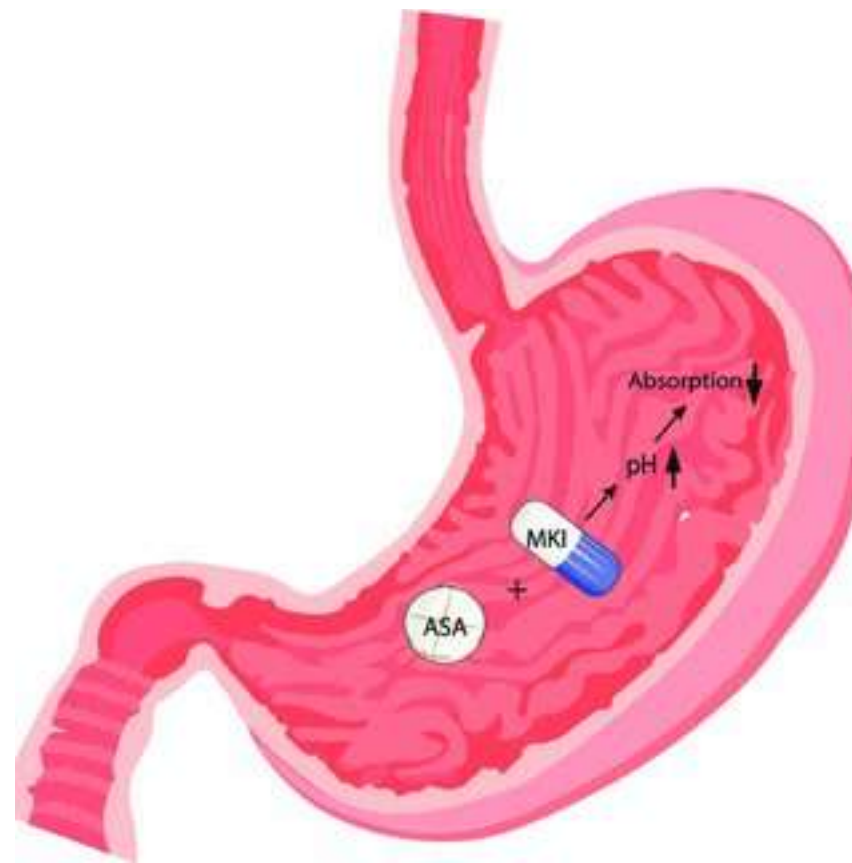
1- Interactions
at the site of
absorption

1- Gastric
motility

2- GIT PH
changes:

3- Binding or
chelation:

4- Structure
of gut wall:



1- Interactions at the site of absorption

1- Gastric motility

2- GIT PH changes:

3- Binding or chelation:

4- Structure of gut wall:

3- Binding or chelation:

A- Many drugs interfere with absorption of Tetracyclines and ciprofloxacin:

1- Mg^{++} , Al^{+++} and Ca^{++} antacids and ferrous salts chelate tetracyclines and ciprofloxacin.

2- dietary supplements containing iron or zinc.

3- products containing Ca^{++} (milk, cheese and yoghurt)

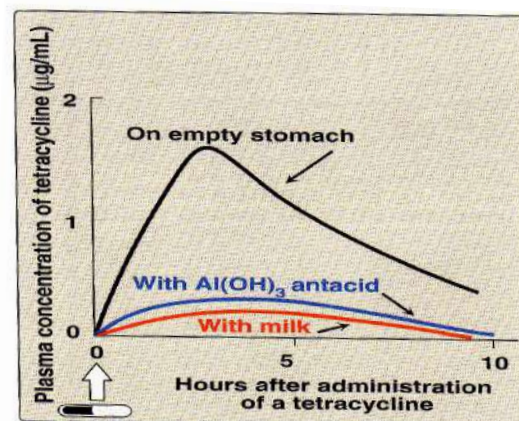


Figure 32.5
Effect of antacids and milk on the absorption of tetracyclines.

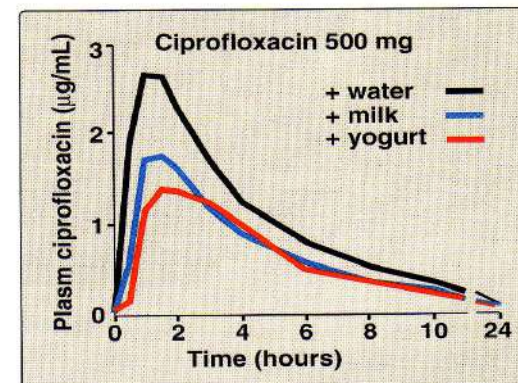


Figure 33.6
Effect of dietary calcium on the absorption of ciprofloxacin.

B- Cholestyramine interfere with absorption of warfarin, digitalis and thyroxine.

C- Activated charcoal interfere with absorption of most of drugs.

1- Interactions
at the site of
absorption

1- Gastric
motility

2- GIT PH
changes:

3- Binding or
chelation:

4- Structure
of gut wall:

4- Structure of gut wall:

A- Phenytoin interferes with absorption of folic acid.

B- Para-amino-salicylic acid and colchicine interfere with absorption of vit B12.

A) Pharmacokinetic interactions (ADME)

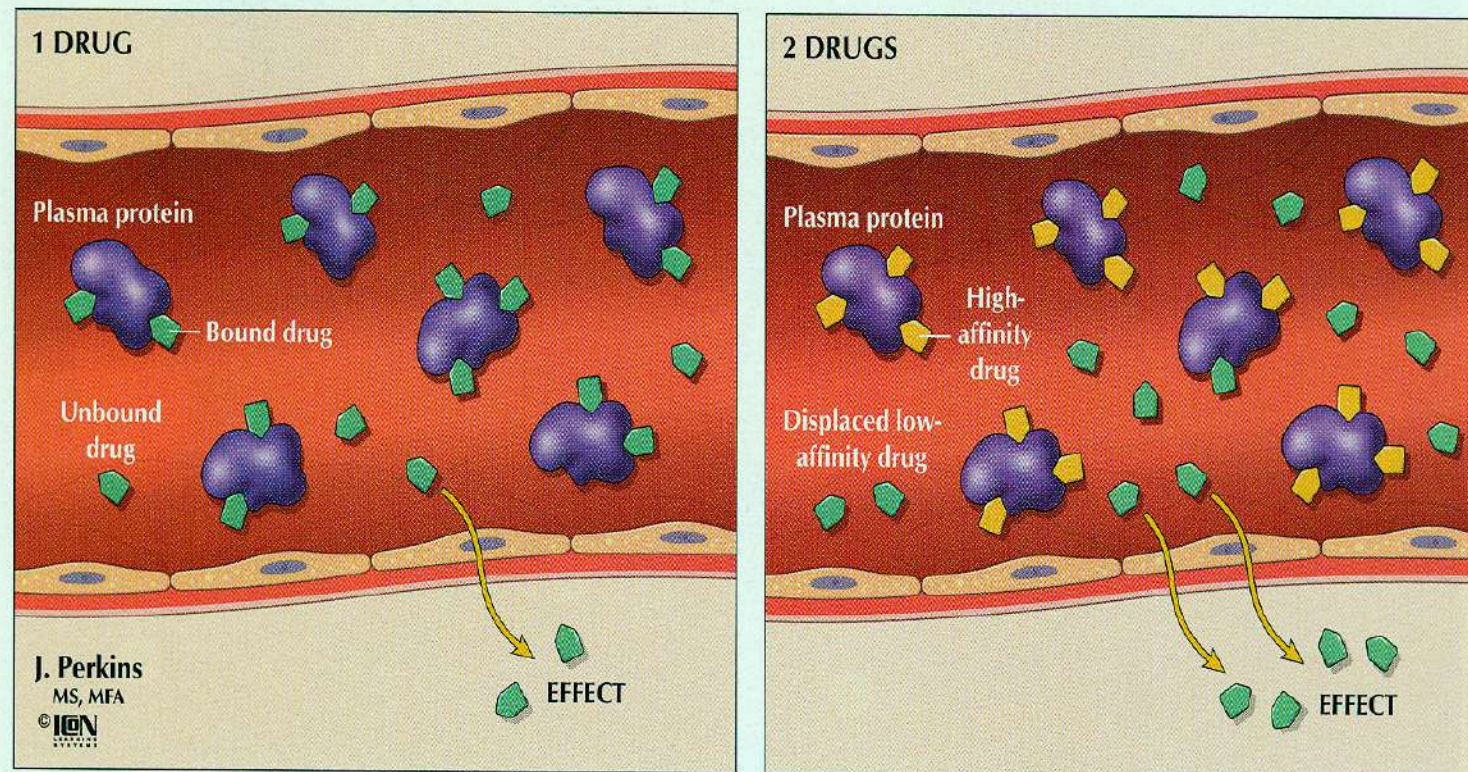
2-Interactions
during
distribution

-Displacement
from plasma
proteins

2- Plasma protein binding sites

- a- Aspirin with Warfarin leading to bleeding
- b- Sulfonamides/ Tolbutamide leading to hypoglycemia.
- c- Bilirubin leading to hyperbilirubinemia in adults and kernicterus in premature infants.

N.B aspirin & sulfonamides displace other drugs from plasma proteins



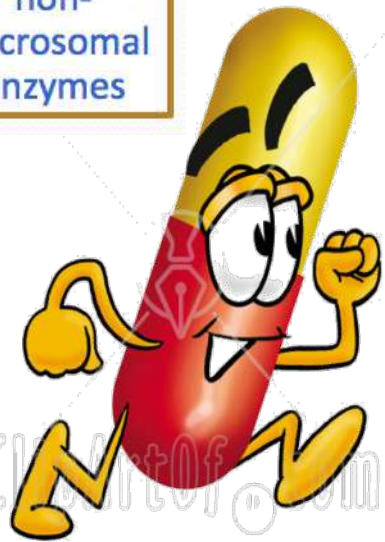
A) Pharmacokinetic interactions (ADME)

3-Interactions during metabolism

- Enzyme inhibition

- Enzyme induction

Inhibition of non-microsomal enzymes



Drug A

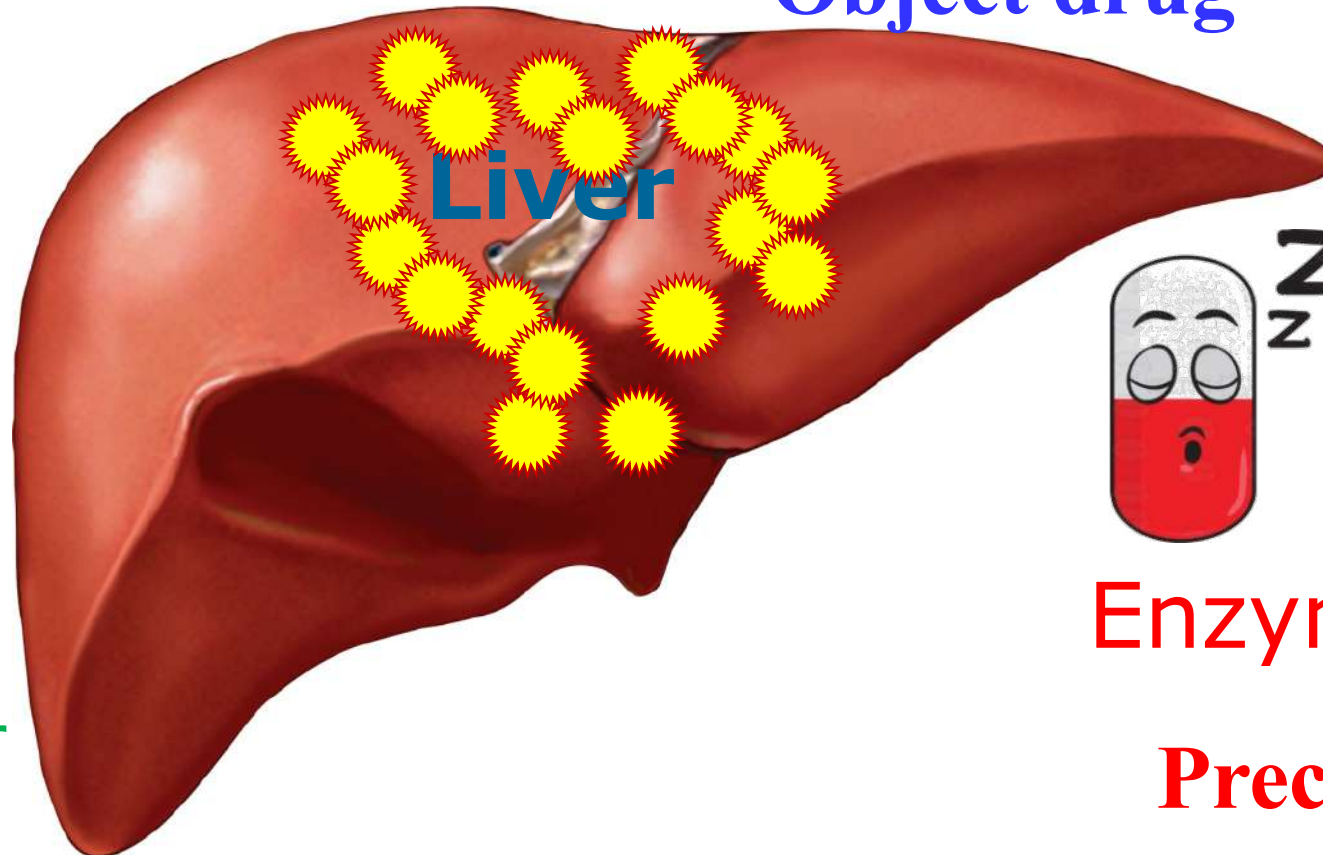
Enzyme Inducer



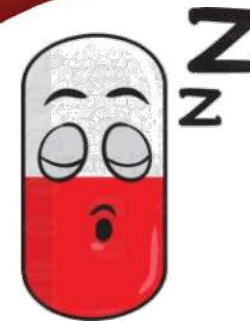
Drug B

Metabolized by liver

Object drug



Liver



Drug C

Enzyme Inhibitor

Precipitant drug

This increases the amount of Drug B in the body and could lead to an overdose or toxic effects

Precipitant drug

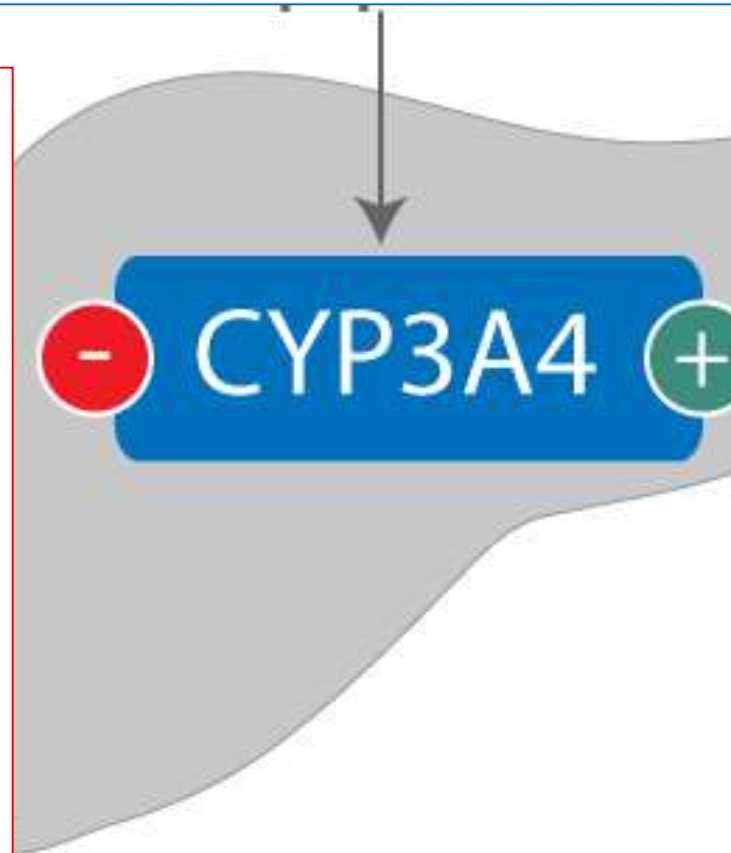
This reduces the amount of Drug B and may lead to loss of Drug B's effectiveness

A) Pharmacokinetic interactions (ADME)

3- Biotransformation

Drugs that metabolized by hepatic CYP

Oral contraceptive.
Warfarin.



Enzyme Inducers

Smoking.
Ethanol
Barbiturates (Anti epileptic)
Carbamazepine (Anti epileptic)
Phenytoin (Anti epileptic)
Rifampin (Antibiotic)
Dexamethasone

Enzyme Inhibitors

Cimitidine (H2 bloker)
Erythromycin (Antibiotic)
Grapefruit juice
Ketoconazole (Antifungal)
Ciprofloxacin (Antibiotic)
Quinidine (Antiarrhythmia)
Ritonavir (AntiHIV)
Sulfonamides (Antimicrobial)
Omeprazole (Antacid)

3-Interactions during metabolism

- Enzyme inhibition

- Enzyme induction

Inhibition of non-microsomal enzymes

A) Pharmacokinetic interactions (ADME)

3- Biotransformation

3-Interactions during metabolism

- Enzyme inhibition

- Enzyme induction

Inhibition of non-microsomal enzymes

inhibitors:

1- Warfarin / azole antifungals

Imidazole (Ketoconazole) antifungals are known to **inhibit** cytochrome P450 enzyme and potentiate action of warfarin.

2- Warfarin / Ciprofloxacin

Ciprofloxacin is known to **inhibit** cytochrome P450 and potentiate action of warfarin

Inducers :

1-Contraceptive Pills/Rifampin

Rifampin is a potent **inducer** of the cytochrome isoenzymes CYP2C and CYP3A which are involved in the metabolism of oral contraceptives.

Concomitant administration of rifampin and oral contraceptives can lead to contraceptive failure.

2-Contraceptive Pills / Phenytoin

Phenytoin is an enzyme **inducer** that increased the activity of cytochrome P 450 and increased the metabolism of components of the oral contraceptive pills, so decreased estrogen levels and decreased estrogen effectiveness and increased the risk of pregnancy

A) Pharmacokinetic interactions (ADME)

3- Biotransformation

3-Interactions during metabolism

- Enzyme inhibition

- Enzyme induction

Inhibition of non-microsomal enzymes

C- Inhibition of non-microsomal enzymes

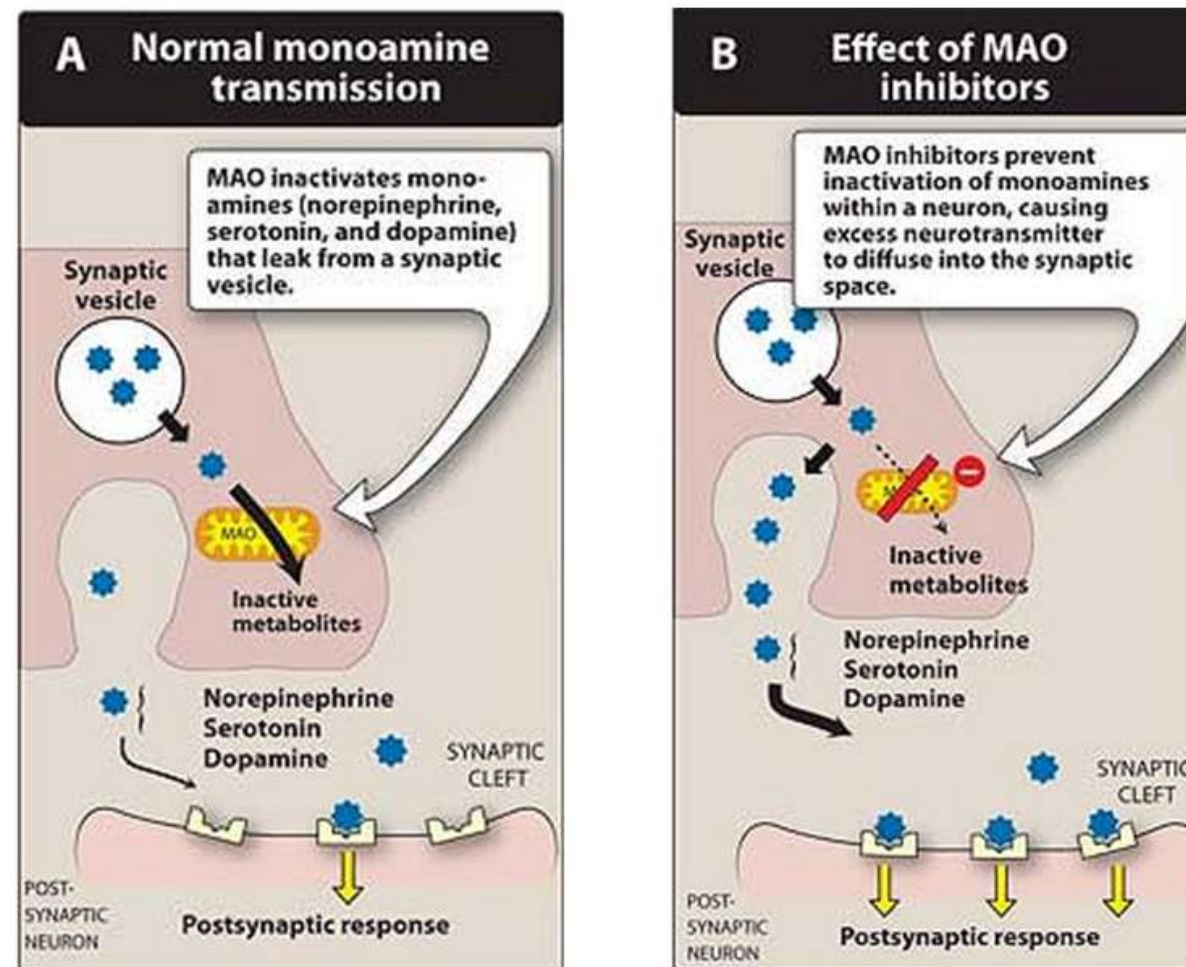
1- (MAOI/SSRI)

- Drugs that can inhibit **MAO**, such as the antidepressant **tranylcypromine**, can retard the metabolism of various drugs, such as **barbiturates and benzodiazepines**, and of **amines, such as serotonin and norepinephrine**. All selective serotonin reuptake inhibitors, including paroxetine, can cause a life-threatening reaction characterized by hyperthermia, rigidity, and coma if administered concurrently with an MAO inhibitor.

2- (6-mercaptopurine / allopurinol)

- Because **6-mercaptopurine** is inactivated in part by **xanthine oxidase**, the concurrent administration of the xanthine oxidase inhibitor, **allopurinol**, can result in an accumulation of 6-mercaptopurine with a consequent increase in bone marrow depression.

- **3- MAOI non-selective / Tyramine**
- Cheese contain tyramine (indirect sympathomimetic) which is normally inactivated by MAO in the gut.
- Individuals taking MAOI are unable to degrade tyramine obtained from diet.
- Tyramine causes the release of large amounts of stored catecholamines from nerve terminals resulting in toxic threshold.



A) Pharmacokinetic interactions (ADME)

4- Renal excretion

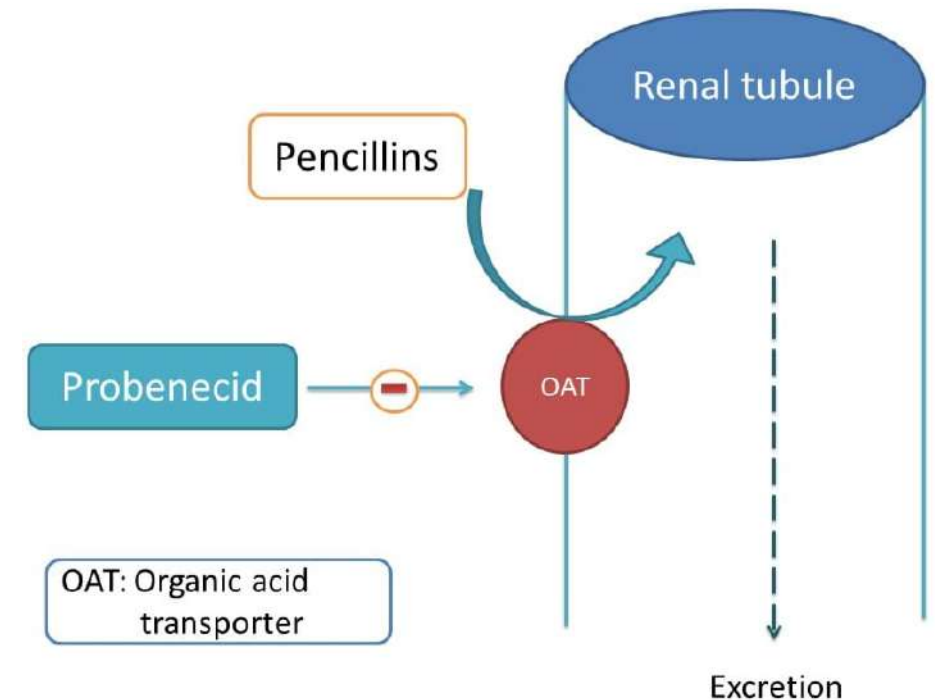
4- interaction during **excretion**

A- Competition for active tubular excretion.

B- Changes in urinary PH

A- Competition for active tubular excretion:

1- **Probenecid** interferes with excretion of **penicillin**, methotrexate, and indomethacin so prolongs their action.



2- **Aspirin** interferes with excretion of **methotrexate** and can cause serious adverse effects.
(Aspirin can also block excretion of methotrexate from kidney. Aspirin has a high displacing capacity , so when given together with methotrexate displaces methotrexate from albumin causing increase the level of methotrexate with severe toxicity).

A) Pharmacokinetic interactions (ADME)

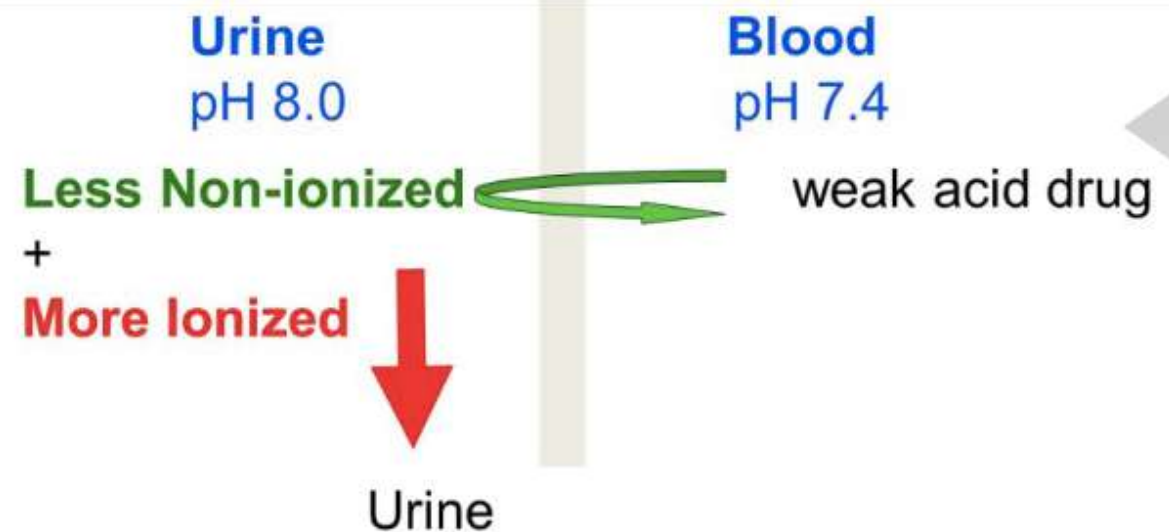
4- Renal excretion

-B- Changes in urinary PH:

- **Acidification** of urine (e.g. ammonium chloride) increases excretion of weak bases e.g amphetamine.
- **Alkalinization** of urine (e.g. sodium bicarbonate) increases excretion of weak acids e.g. Aspirin, phenobarbital and sulfa

This process is called ion trapping.

In presence of sodium bicarbonate, urine is **alkaline** and **more** excretion of acidic drug into urine

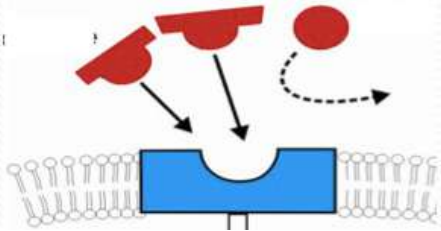


Most of **acidic drug** will be **eliminated** into **alkaline urine**.

Pharmacodynamics

Interactions directly on:

- Receptors
- Body systems.



Pharmacodynamic interactions

- Pharmacodynamic interactions generally involve additive, synergistic or antagonistic effects of drugs acting on the same receptors or physiological systems. These interactions are much less easy to classify than those with a pharmacokinetic basis.



	Interacting drugs	Pharmacological effect
1-	NSAID + ACE inhibitors	NSAIDs cause Na ⁺ retention and decrease effect of ACE inhibitors
2-	NSAIDs + warfarin	Increase risk of bleeding
	ACE inhibitors + K ⁺ sparing diuretics	Increase risk of hyperkalemia
4-	Verapamil + β blockers	Bradycardia and asystole
5-	Neuromuscular blockers + aminoglycosides	Increase neuromuscular blockade.
6-	Telithromycin + Amiodarone (Class III antiarrhythmic)	Increase risk of QT interval prolongation
7-	Ganciclovir+ Zidovudine	Increase risk of bone marrow suppression